

问题1:
曲边梯形的面积

问题2:
变速直线运动的路程

存在定理

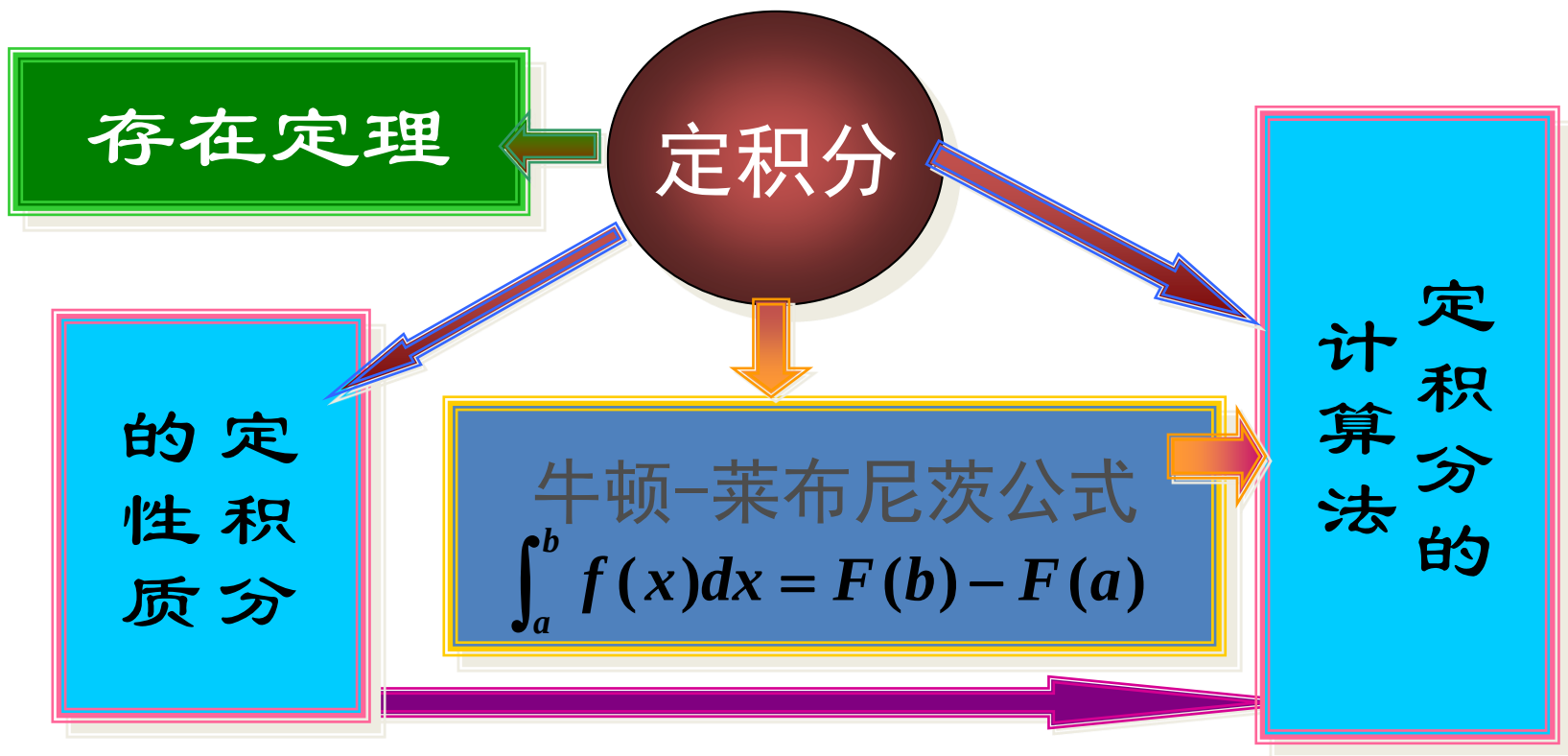
定积分

定积分
的性质

牛顿-莱布尼茨公式

$$\int_a^b f(x)dx = F(b) - F(a)$$

定积分的
算法



原 函 数

不 定 积 分

分部
积分法

积分法

直接
积分法

基本
积分
表

第一换元法
第二换元法

几种特殊类
函数的积分

§ 6.2 定积分的基本积分法

- 定积分的换元法
- 定积分的分部积分法

一、定积分的第一类换元积分法(凑微分法)

例1 计算 $\int_0^{\frac{\pi}{2}} \cos^5 x \sin x dx$.

解 $\int_0^{\frac{\pi}{2}} \cos^5 x \sin x dx = -\int_0^{\frac{\pi}{2}} \cos^5 x d(\cos x)$

$$= -\int_0^{\frac{\pi}{2}} \cos^5 x d \cos x$$

$$= -\left. \frac{\cos^6 x}{6} \right|_0^{\frac{\pi}{2}} = \frac{1}{6}.$$

$$\cos x = u$$

$$= -\int_1^0 u^5 du$$

$$= -\left. \frac{u^6}{6} \right|_1^0 = \frac{1}{6}$$

注意：
换元必换限

定理4 设 $u = g(x)$, 如果 $g'(x)$ 在 $[a, b]$ 上连续,
 $f(u)$ 在 $g(x)$ 的值域区间 I 上连续, 则

$$\begin{aligned}\int_a^b f(g(x))g'(x)dx &= \int_a^b f(g(x))d(g(x)) \\ &= \int_{g(a)}^{g(b)} f(u)du\end{aligned}$$

注意：
换元必换限

例2 计算 $\int_0^{\pi} \sqrt{\sin^3 x - \sin^5 x} dx$.

解 $\because f(x) = \sqrt{\sin^3 x - \sin^5 x} = |\cos x|(\sin x)^{\frac{3}{2}}$

$$\therefore \int_0^{\pi} \sqrt{\sin^3 x - \sin^5 x} dx = \int_0^{\pi} |\cos x|(\sin x)^{\frac{3}{2}} dx$$

$$= \int_0^{\frac{\pi}{2}} \cos x (\sin x)^{\frac{3}{2}} dx - \int_{\frac{\pi}{2}}^{\pi} \cos x (\sin x)^{\frac{3}{2}} dx$$

$$= \int_0^{\frac{\pi}{2}} (\sin x)^{\frac{3}{2}} d \sin x - \int_{\frac{\pi}{2}}^{\pi} (\sin x)^{\frac{3}{2}} d \sin x$$

$$= \frac{2}{5} (\sin x)^{\frac{5}{2}} \Big|_0^{\frac{\pi}{2}} - \frac{2}{5} (\sin x)^{\frac{5}{2}} \Big|_{\frac{\pi}{2}}^{\pi} = \frac{4}{5}.$$

例3 计算 $\int_{\sqrt{e}}^{e^{\frac{3}{4}}} \frac{dx}{x \sqrt{\ln x(1-\ln x)}}$.

解 原式 $= \int_{\sqrt{e}}^{e^{\frac{3}{4}}} \frac{d(\ln x)}{\sqrt{\ln x(1-\ln x)}}$

$$= \int_{\sqrt{e}}^{e^{\frac{3}{4}}} \frac{d(\ln x)}{\sqrt{\ln x} \sqrt{(1-\ln x)}} = 2 \int_{\sqrt{e}}^{e^{\frac{3}{4}}} \frac{d \sqrt{\ln x}}{\sqrt{1-(\sqrt{\ln x})^2}}$$

$$= 2 \left[\arcsin(\sqrt{\ln x}) \right]_{\sqrt{e}}^{e^{\frac{3}{4}}} = \frac{\pi}{6}.$$

二、定积分的第二类换元积分法

定理5 设 $f(x)$ 在 $[a, b]$ 上连续, 若变换 $x = g(t)$ 满足

(1) $g(t)$ 在闭区间 $[\alpha, \beta]$ (或 $[\beta, \alpha]$) 上有连续导数 $g'(t)$;

(2) 当 $t \in [\alpha, \beta]$ (或 $[\beta, \alpha]$) 时, $a \leq g(t) \leq b$

(3) $g(\alpha) = a$ 、 $g(\beta) = b$,

则有下面的公式成立

$$\int_a^b f(x) dx = \int_{\alpha}^{\beta} f[g(t)] g'(t) dt .$$

注意 当 $\alpha > \beta$ 时, 换元公式仍成立.

应用换元公式时应注意:

- (1) 用 $x = g(t)$ 把变量 x 换成新变量 t 时, 积分限也相应的改变.
- (2) 求出 $f[g(t)]g'(t)$ 的一个原函数 $\Phi(t)$ 后, 不必象计算不定积分那样再要把 $\Phi(t)$ 变换成原变量 x 的函数, 而只要把新变量 t 的上、下限分别代入 $\Phi(t)$ 然后相减就行了.

例3. 计算 $\int_0^4 \frac{dx}{1 + \sqrt{x}}$

解: 令 $x = t^2$, 则 $dx = 2t dt$

当 $x = 0$ 时, $t = 0$; 当 $x = 4$ 时, $t = 2$;

$$\begin{aligned} \int_0^4 \frac{dx}{1 + \sqrt{x}} &= \int_0^2 \frac{2t}{1 + t} dt \\ &= 2 \int_0^2 \left(1 - \frac{1}{1 + t}\right) dt = 2 \left[t - \ln(1 + t) \right] \Big|_0^2 \\ &= 2(2 - \ln 3) \end{aligned}$$

例5 计算 $\int_0^a \frac{1}{x + \sqrt{a^2 - x^2}} dx$. ($a > 0$)

解 令 $x = a \sin t$, $dx = a \cos t dt$,

$$x = a \Rightarrow t = \frac{\pi}{2}, \quad x = 0 \Rightarrow t = 0,$$

$$\begin{aligned} \text{原式} &= \int_0^{\frac{\pi}{2}} \frac{a \cos t}{a \sin t + \sqrt{a^2 (1 - \sin^2 t)}} dt \\ &= \int_0^{\frac{\pi}{2}} \frac{\cos t}{\sin t + \cos t} dt = \frac{1}{2} \int_0^{\frac{\pi}{2}} \left(1 + \frac{\cos t - \sin t}{\sin t + \cos t} \right) dt \\ &= \frac{1}{2} \cdot \frac{\pi}{2} + \frac{1}{2} \left[\ln |\sin t + \cos t| \right]_0^{\frac{\pi}{2}} = \frac{\pi}{4}. \end{aligned}$$

$$\text{例4 } \int_{-2}^{-\sqrt{2}} \frac{dx}{x\sqrt{x^2-1}} \quad \underline{\underline{x = \sec t}}$$

$$\int_{\frac{2\pi}{3}}^{\frac{3\pi}{4}} \frac{1}{\sec t \cdot |\tan t|} \sec t \cdot \tan t dt$$

$$= -\int_{\frac{2\pi}{3}}^{\frac{3\pi}{4}} dt = -\frac{\pi}{12}.$$

定理 6 当 $f(x)$ 在 $[-l, l]$ 上连续, 且有

① $f(x)$ 为偶函数, 则

$$\int_{-l}^l f(x)dx = 2\int_0^l f(x)dx;$$

② $f(x)$ 为奇函数, 则 $\int_{-l}^l f(x)dx = 0$.

证
$$\int_{-l}^l f(x)dx = \int_{-l}^0 f(x)dx + \int_0^l f(x)dx,$$

在 $\int_{-l}^0 f(x)dx$ 中令 $x = -t$,

$$\int_{-l}^0 f(x)dx = -\int_l^0 f(-t)dt = \int_0^l f(-t)dt,$$

① $f(x)$ 为偶函数, 则 $f(-t) = f(t)$,

$$\begin{aligned}\int_{-l}^l f(x)dx &= \int_{-l}^0 f(x)dx + \int_0^l f(x)dx \\ &= 2\int_0^l f(x)dx;\end{aligned}$$

② $f(x)$ 为奇函数, 则 $f(-t) = -f(t)$,

$$\int_{-l}^l f(x)dx = \int_{-l}^0 f(x)dx + \int_0^l f(x)dx = 0.$$

例5 计算 $\int_{-1}^1 \frac{2x^2 + x \cos x}{1 + \sqrt{1-x^2}} dx$.

解 原式 = $\int_{-1}^1 \frac{2x^2}{1 + \sqrt{1-x^2}} dx + \int_{-1}^1 \frac{x \cos x}{1 + \sqrt{1-x^2}} dx$

偶函数 奇函数

$$= 4 \int_0^1 \frac{x^2}{1 + \sqrt{1-x^2}} dx = 4 \int_0^1 \frac{x^2(1 - \sqrt{1-x^2})}{1 - (1-x^2)} dx$$

$$= 4 \int_0^1 (1 - \sqrt{1-x^2}) dx = 4 - 4 \int_0^1 \sqrt{1-x^2} dx$$

单位圆的面积

$$= 4 - \pi.$$

P310 例10

例 6 若 $f(x)$ 在 $[0,1]$ 上连续, 证明

$$(1) \int_0^{\frac{\pi}{2}} f(\sin x) dx = \int_0^{\frac{\pi}{2}} f(\cos x) dx;$$

$$(2) \int_0^{\pi} x f(\sin x) dx = \frac{\pi}{2} \int_0^{\pi} f(\sin x) dx.$$

由此计算 $\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx.$

证 (1) $x = \frac{\pi}{2} - t \Rightarrow dx = -dt,$
设

$$x = 0 \Rightarrow t = \frac{\pi}{2}, \quad x = \frac{\pi}{2} \Rightarrow t = 0,$$

$$\begin{aligned}\int_0^{\frac{\pi}{2}} f(\sin x) dx &= -\int_{\frac{\pi}{2}}^0 f\left[\sin\left(\frac{\pi}{2}-t\right)\right] dt \\ &= \int_0^{\frac{\pi}{2}} f(\cos t) dt = \int_0^{\frac{\pi}{2}} f(\cos x) dx;\end{aligned}$$

同理可证明: $\int_0^{\frac{\pi}{2}} f^n(\sin x) dx = \int_0^{\frac{\pi}{2}} f^n(\cos x) dx;$

(2) 设 $x = \pi - t \Rightarrow dx = -dt,$

$$x = 0 \Rightarrow t = \pi, \quad x = \pi \Rightarrow t = 0,$$

$$\begin{aligned}\int_0^{\pi} x f(\sin x) dx &= -\int_{\pi}^0 (\pi - t) f[\sin(\pi - t)] dt \\ &= \int_0^{\pi} (\pi - t) f(\sin t) dt,\end{aligned}$$

$$\begin{aligned}
 \int_0^{\pi} x f(\sin x) dx &= \pi \int_0^{\pi} f(\sin t) dt - \int_0^{\pi} t f(\sin t) dt \\
 &= \pi \int_0^{\pi} f(\sin x) dx - \int_0^{\pi} x f(\sin x) dx, \\
 \therefore \int_0^{\pi} x f(\sin x) dx &= \frac{\pi}{2} \int_0^{\pi} f(\sin x) dx.
 \end{aligned}$$

$$\begin{aligned}
 \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx &= \frac{\pi}{2} \int_0^{\pi} \frac{\sin x}{1 + \cos^2 x} dx \\
 &= -\frac{\pi}{2} \int_0^{\pi} \frac{1}{1 + \cos^2 x} d(\cos x) = -\frac{\pi}{2} [\arctan(\cos x)]_0^{\pi} \\
 &= -\frac{\pi}{2} \left(-\frac{\pi}{4} - \frac{\pi}{4} \right) = \frac{\pi^2}{4}.
 \end{aligned}$$

例7 证明: $\int_x^1 \frac{1}{1+x^2} dx = \int_1^{\frac{1}{x}} \frac{1}{1+x^2} dx \quad (x > 0).$

证明:先区分自变量与积分变量.

即证: $\int_x^1 \frac{1}{1+t^2} dt = \int_1^{\frac{1}{x}} \frac{1}{1+t^2} dt$

$$\int_x^1 \frac{1}{1+t^2} dt \quad \underline{\underline{t = u^{-1}, u = t^{-1}}} \quad \int_{\frac{1}{x}}^1 \frac{-\frac{1}{u^2}}{1 + \frac{1}{u^2}} du = -\int_{\frac{1}{x}}^1 \frac{1}{1+u^2} du$$

$$= \int_1^{\frac{1}{x}} \frac{1}{1+u^2} du = \int_1^{\frac{1}{x}} \frac{1}{1+t^2} dt$$

定理7 设 $f(x)$ 在实轴上连续且是以 T 为周期的周期函数,则对任意实数 a 成立

$$\int_a^{a+T} f(x)dx = \int_0^T f(x)dx$$

证明: $\int_a^{a+T} f(x)dx = \int_a^0 f(x)dx + \int_0^T f(x)dx + \int_T^{a+T} f(x)dx$

$$\int_T^{a+T} f(x)dx = \int_0^a f(t+T)dt \quad (x = t+T)$$

$$= \int_0^a f(t)dt = \int_0^a f(x)dx$$

所以 $\int_T^{a+T} f(x)dx + \int_a^0 f(x)dx = \int_0^a f(x)dx + \int_a^0 f(x)dx$
 $= 0$

即 $\int_a^{a+T} f(x)dx = \int_0^T f(x)dx$

定理 7 表明:

- 周期为 T 的连续函数, 在任一长度为 T 的闭区间上的定积分均相等. 即与积分区间的起点无关.

推论: 若函数 $f(x)$ 是以 T 为周期的连续函数, 则

$$\int_0^{nT} f(x)dx = n \int_0^T f(x)dx$$

二、定积分的分部积分法

设函数 $f(x)$ 、 $g(x)$ 在区间 $[a, b]$ 上具有连续导数，则有

$$\int_a^b f(x)dg(x) = \left. f(x)g(x) \right|_a^b - \int_a^b g(x)df(x).$$

定积分的分部积分公式

例8 计算 $\int_0^{\frac{1}{2}} \arcsin x dx$.

解 令 $f(x) = \arcsin x$, $dg(x) = dx$,

则 $df(x) = \frac{dx}{\sqrt{1-x^2}}$, $g(x) = x$,

$$\int_0^{\frac{1}{2}} \arcsin x dx = \left[x \arcsin x \right]_0^{\frac{1}{2}} - \int_0^{\frac{1}{2}} \frac{x dx}{\sqrt{1-x^2}}$$

$$= \frac{1}{2} \cdot \frac{\pi}{6} + \frac{1}{2} \int_0^{\frac{1}{2}} \frac{1}{\sqrt{1-x^2}} d(1-x^2)$$

$$= \frac{\pi}{12} + \left[\sqrt{1-x^2} \right]_0^{\frac{1}{2}} = \frac{\pi}{12} + \frac{\sqrt{3}}{2} - 1.$$

例9 计算 $\int_0^{\frac{\pi}{4}} \frac{x dx}{1 + \cos 2x}$.

解 $\because 1 + \cos 2x = 2 \cos^2 x,$

$$\therefore \int_0^{\frac{\pi}{4}} \frac{x dx}{1 + \cos 2x} = \int_0^{\frac{\pi}{4}} \frac{x dx}{2 \cos^2 x} = \int_0^{\frac{\pi}{4}} \frac{x}{2} d(\tan x)$$

$$= \frac{1}{2} [x \tan x]_0^{\frac{\pi}{4}} - \frac{1}{2} \int_0^{\frac{\pi}{4}} \tan x dx$$

$$= \frac{\pi}{8} - \frac{1}{2} [\ln \sec x]_0^{\frac{\pi}{4}} = \frac{\pi}{8} - \frac{\ln 2}{4}.$$

例10 计算 $\int_0^1 \frac{\ln(1+x)}{(2+x)^2} dx$.

解 $\int_0^1 \frac{\ln(1+x)}{(2+x)^2} dx = -\int_0^1 \ln(1+x) d \frac{1}{2+x}$

$$= -\left[\frac{\ln(1+x)}{2+x} \right]_0^1 + \int_0^1 \frac{1}{2+x} d \ln(1+x)$$

$$= -\frac{\ln 2}{3} + \int_0^1 \boxed{\frac{1}{2+x} \cdot \frac{1}{1+x}} dx \rightarrow \frac{1}{1+x} - \frac{1}{2+x}$$

$$= -\frac{\ln 2}{3} + [\ln(1+x) - \ln(2+x)]_0^1 = \frac{5}{3} \ln 2 - \ln 3.$$

例11 求定积分 $\int_1^{e^\pi} \cos(\ln x) dx$.

解 $\int_1^{e^\pi} \cos(\ln x) dx = [x \cos(\ln x)]_1^{e^\pi} - \int_1^{e^\pi} x d[\cos(\ln x)]$

$$= -e^\pi - 1 + \int_1^{e^\pi} x \sin(\ln x) \cdot \frac{1}{x} dx$$

$$= -e^\pi - 1 + [x \sin(\ln x)]_1^{e^\pi} - \int_1^{e^\pi} x d[\sin(\ln x)]$$

$$= -e^\pi - 1 - \int_1^{e^\pi} \cos(\ln x) dx$$

$$\therefore \int_1^{e^\pi} \cos(\ln x) dx = -\frac{e^\pi + 1}{2}$$

例13 证明定积分公式

$$I_n = \int_0^{\frac{\pi}{2}} \sin^n x dx = \int_0^{\frac{\pi}{2}} \cos^n x dx$$
$$= \begin{cases} \frac{n-1}{n} \cdot \frac{n-3}{n-2} \cdots \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2}, & n \text{ 为正偶数} \\ \frac{n-1}{n} \cdot \frac{n-3}{n-2} \cdots \frac{4}{5} \cdot \frac{2}{3}, & n \text{ 为大于1的正奇数} \end{cases}$$

证 设 $f(x) = \sin^{n-1} x$, $dg(x) = \sin x dx$,

$$g(x) = -\cos x,$$

$$I_n = \left[\underbrace{-\sin^{n-1} x \cos x}_0 \right]_0^{\frac{\pi}{2}} + (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x \underbrace{\cos^2 x}_{1-\sin^2 x} dx$$

$$I_n = (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x dx - (n-1) \int_0^{\frac{\pi}{2}} \sin^n x dx$$

$$= (n-1)I_{n-2} - (n-1)I_n$$

$$I_n = \frac{n-1}{n} I_{n-2} \quad \text{积分 } I_n \text{ 关于下标的递推公式}$$

$$I_{n-2} = \frac{n-3}{n-2} I_{n-4} \quad \dots\dots, \text{直到下标减到0或1为止}$$

$$I_{2m} = \frac{2m-1}{2m} \cdot \frac{2m-3}{2m-2} \cdots \frac{5}{6} \cdot \frac{3}{4} \cdot \frac{1}{2} I_0, \quad (m=1,2,\cdots)$$

$$I_{2m+1} = \frac{2m}{2m+1} \cdot \frac{2m-2}{2m-1} \cdots \frac{6}{7} \cdot \frac{4}{5} \cdot \frac{2}{3} I_1,$$

$$I_0 = \int_0^{\frac{\pi}{2}} dx = \frac{\pi}{2}, \quad I_1 = \int_0^{\frac{\pi}{2}} \sin x dx = 1,$$

于是
$$I_{2m} = \frac{2m-1}{2m} \cdot \frac{2m-3}{2m-2} \cdots \frac{5}{6} \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2},$$

$$I_{2m+1} = \frac{2m}{2m+1} \cdot \frac{2m-2}{2m-1} \cdots \frac{6}{7} \cdot \frac{4}{5} \cdot \frac{2}{3}.$$

例14 求定积分 $\int_0^{\frac{\pi}{2}} \sin^7 x dx$

$$\text{解 } \int_0^{\frac{\pi}{2}} \sin^7 x dx = \frac{6}{7} \cdot \frac{4}{5} \cdot \frac{2}{3} = \frac{16}{35}$$

例15 求定积分 $\int_0^{\frac{\pi}{2}} \cos^8 x dx$

$$\text{解 } \int_0^{\frac{\pi}{2}} \cos^8 x dx = \frac{7}{8} \cdot \frac{5}{6} \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} = \frac{35\pi}{256}$$

$$\text{又例 } \int_0^{\frac{\pi}{2}} \cos^8 x \sin^2 x dx = \int_0^{\frac{\pi}{2}} \cos^8 x (1 - \cos^2 x) dx$$

$$= \left(\frac{7 \times 5 \times 3 \times 1}{8 \times 6 \times 4 \times 2} - \frac{9 \times 7 \times 5 \times 3 \times 1}{10 \times 8 \times 6 \times 4 \times 2} \right) \frac{\pi}{2}$$

$$= \left(1 - \frac{9}{10} \right) \frac{7 \times 5 \times 3 \times 1}{8 \times 6 \times 4 \times 2} \frac{\pi}{2} = \frac{7\pi}{512}.$$

例12 设 $f(x) = \int_1^{x^2} \frac{\sin t}{t} dt$, 求 $\int_0^1 xf(x)dx$.

解 因为 $\frac{\sin t}{t}$ 没有初等形式的原函数,
无法直接求出 $f(x)$, 所以采用分部积分法

$$\int_0^1 xf(x)dx = \frac{1}{2} \int_0^1 f(x) d(x^2)$$

$$= \frac{1}{2} [x^2 f(x)]_0^1 - \frac{1}{2} \int_0^1 x^2 df(x)$$

$$= \frac{1}{2} f(1) - \frac{1}{2} \int_0^1 x^2 f'(x) dx$$

$$\because f(x) = \int_1^{x^2} \frac{\sin t}{t} dt, \quad f(1) = \int_1^1 \frac{\sin t}{t} dt = 0,$$

$$f'(x) = \frac{\sin x^2}{x^2} \cdot 2x = \frac{2\sin x^2}{x},$$

$$\therefore \int_0^1 xf(x)dx = \frac{1}{2} f(1) - \frac{1}{2} \int_0^1 x^2 f'(x)dx$$

$$= -\frac{1}{2} \int_0^1 2x \sin x^2 dx = -\frac{1}{2} \int_0^1 \sin x^2 dx^2$$

$$= \frac{1}{2} [\cos x^2]_0^1 = \frac{1}{2} (\cos 1 - 1).$$

例16 已知 $f(x) = \int_0^x e^{\cos t} dt$, 求 $\int_0^\pi f(x) \cos x dx$

解 $\int_0^\pi f(x) \cos x dx = \int_0^\pi f(x) d \sin x$

$$= f(x) \sin x \Big|_0^\pi - \int_0^\pi \sin x f'(x) dx$$

$$= - \int_0^\pi \sin x e^{\cos x} dx$$

$$= \int_0^\pi e^{\cos x} d \cos x$$

$$= e^{\cos x} \Big|_0^\pi$$

$$= e^{-1} - 1$$

例17 证明 $\int_0^x f(u)(x-u)du = \int_0^x \left[\int_0^u f(t)dt \right] du$

证明: $\int_0^x \left[\int_0^u f(t)dt \right] du$

$$= u \int_0^u f(t)dt \Big|_0^x - \int_0^x uf(u)du$$
$$= x \int_0^x f(t)dt - \int_0^x uf(u)du$$
$$= x \int_0^x f(u)du - \int_0^x uf(u)du$$
$$= \int_0^x f(u)(x-u)du$$